

Microsoft Azure

Cloud Computing Notes — Knowledge Base

1. Overview

Microsoft Azure is Microsoft's cloud platform, offering compute, storage, database, AI, and networking services. It has particularly strong integration with Microsoft's enterprise ecosystem (Active Directory, Windows Server, Office 365, .NET).

2. Core Services by Category

Category	Key Services
Compute	Virtual Machines, App Service, Azure Functions, AKS
Storage	Blob Storage, Disk Storage, Files
Database	Azure SQL Database, Cosmos DB, Azure Database for PostgreSQL
Identity	Azure Active Directory (Entra ID), RBAC
DevOps	Azure DevOps, GitHub Actions integration, ARM/Bicep templates

3. Distinctive Strengths

- Deep integration with enterprise Microsoft tools (Active Directory, Office 365, Windows Server) — attractive for organizations already on Microsoft stacks.
- Hybrid cloud support via Azure Arc and Azure Stack, letting on-premise and cloud resources be managed together.
- Cosmos DB offers globally distributed, multi-model database capabilities with tunable consistency levels.
- Strong presence in regulated industries due to compliance certifications.

4. Core Concepts

- Resource Groups — logical containers for related resources, simplifying management and billing.
- Azure Active Directory (Entra ID) — identity and access management, tightly integrated across services.
- ARM Templates / Bicep — infrastructure-as-code for repeatable deployments.

5. Considerations

- Service catalog and naming conventions can differ meaningfully from AWS/GCP, adding a learning curve when switching providers.

- Cost management tools exist (Azure Cost Management) but require active monitoring like other clouds.